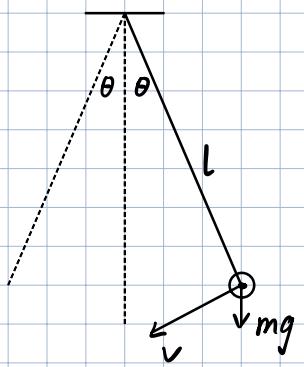


$$ml\frac{d^2\theta}{dt^2} = -mg\sin\theta$$

$$\Leftrightarrow \quad \frac{d^2\theta}{dt^2} + \frac{g}{l}\sin\theta = 0$$

$\omega_0 \equiv \sqrt{g/l}$,

将 $\sin\theta$ Taylor 展开：

$$\sin\theta = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots \approx \theta .$$

$$\therefore \quad \frac{d^2\theta}{dt^2} + \omega_0^2\theta = 0,$$

$$\theta = e^{\lambda t}, \quad \Rightarrow \quad \lambda^2 + \omega_0^2 = 0, \quad \lambda = \pm i\omega_0$$

$$\theta(t) = C_1 e^{i\omega_0 t} + C_2 e^{-i\omega_0 t}$$

$\theta(t) \in R$,

$$C_1^* e^{-i\omega_0 t} + C_2^* e^{+i\omega_0 t} = C_1 e^{i\omega_0 t} + C_2 e^{-i\omega_0 t}$$

$$\Leftrightarrow \quad C_1 = C_2^*, \quad C_1^* = C_2 .$$

∴ 可写为：$C_1 = \frac{P}{2} e^{i\varphi}$, $C_2 = \frac{P}{2} e^{-i\varphi}$.

P 为模，φ 为幅角.

$$\theta(t) = \frac{P}{2}\left[e^{i(\omega_0 t+\varphi)} + e^{-i(\omega_0 t+\varphi)}\right]$$
$$= P\cos(\omega_0 t+\varphi).$$

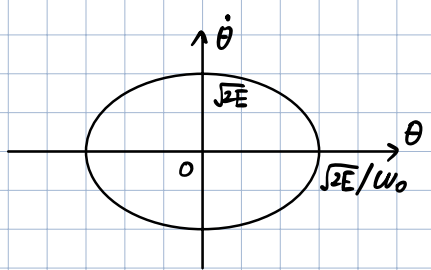
相空间?

对 $\frac{d^2\theta}{dt^2} + \omega_0^2\theta = 0$, 积分:

$$\frac{1}{2}\left(\frac{d\theta}{dt}\right)^2 + \frac{1}{2}\omega_0^2\theta^2 = E.$$

$\Leftrightarrow$
$$\left(\frac{\omega_0}{\sqrt{2E}}\right)^2\theta^2 + \left(\frac{1}{\sqrt{2E}}\right)^2\dot{\theta}^2 = 1.$$

椭圆.



这就是相平面. $\dot{\theta}-\theta$.

上述 Taylor 展开取近似要求 θ 在零点附近. 无限制?

$$\frac{d^2\theta}{dt^2} + \omega_0^2 \sin\theta = 0.$$

$$\frac{d^2\theta}{dt}\frac{d\theta}{dt} + \omega_0^2 \sin\theta \frac{d\theta}{dt} = 0.$$

$$\int \frac{d^2\theta}{dt^2}\frac{d\theta}{dt}\,dt + \int \omega_0^2 \sin\theta \frac{d\theta}{dt}\,dt = 0.$$

$$\int \frac{d\theta}{dt}\,d\frac{d\theta}{dt} + \int \omega_0^2 \sin\theta\, d\theta = 0$$

$$\therefore \quad \left(\frac{d\theta}{dt}\right)^2 = E + 2\omega_0^2 \cos\theta$$

最大位移角? $\frac{d\theta}{dt} = 0$. $E = -2\omega_0^2 \cos\theta_0$.

$$\frac{d\theta}{dt} = \omega_0 [2(\cos\theta - \cos\theta_0)]^{1/2}$$

$$\omega_0 T = \int \frac{d\theta}{\sqrt{2(\cos\theta - \cos\theta_0)}}$$

设 $t=0$ 时 $\theta = 0$, 振动周期为 T, 当 $t = \frac{T}{4}$ 时, $\theta = \theta_0$.

$$\therefore \quad \omega_0 T/4 = \int_0^{\theta_0} \frac{d\theta}{2\sqrt{\sin^2\frac{\theta_0}{2} - \sin^2\frac{\theta}{2}}}$$

令 $\sin\frac{\theta}{2} = \sin\frac{\theta_0}{2}\sin\phi$, 有: $\frac{1}{2}\cos\frac{\theta}{2}\delta\phi = \frac{1}{2}\sin\frac{\theta_0}{2}\cos\phi\,\delta\phi$.

$$\therefore \quad \frac{\pi}{2}\frac{T}{T_0} = \int_0^{\frac{\pi}{2}} \frac{d\phi}{[1 - (\sin^2\theta_0/2)\sin^2\phi]^{1/2}}, \quad T_0 = \frac{2\pi}{\omega_0}$$

Finally,

$$T = T_0\left[1+\left(\frac{1}{2}\right)^2\sin^2\frac{\theta_0}{2}+\left(\frac{1}{2}\cdot\frac{3}{4}\right)^2\sin^4\frac{\theta_0}{2}+\cdots\right].$$

$$\approx T_0\left[1+\frac{1}{4}\sin^2\frac{\theta_0}{2}\right].$$

考虑 $\theta=\pm\pi,\ \dot{\theta}=0$ 附近运动情况. 摆锤倒立.

$$\phi=\pi-\theta.$$

$$\frac{d^2}{dt^2}(\pi-\phi)+\omega_0^2\sin(\pi-\phi)=0.$$

$$\Leftrightarrow \qquad \frac{d^2\phi}{dt^2}-\omega_0^2\sin^2\phi=0.$$

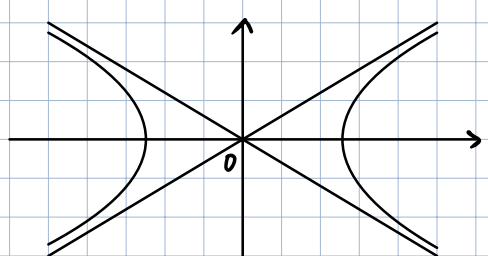
$\sin\phi\approx\phi$,

$$\frac{d^2\phi}{dt^2}-\omega_0^2\phi=0$$

积分:

$$\frac{1}{2}\left(\frac{d\phi}{dt}\right)^2-\frac{1}{2}\omega_0^2\phi^2=0.$$

双曲线.



阻尼振子

阻尼 $\propto \dot{\theta}$

$$ml\frac{d^2\theta}{dt^2} + \gamma l\frac{d\theta}{dt} + mg\sin\theta = 0.$$

$\sin\theta \approx \theta$.

$$\frac{d^2\theta}{dt^2} + 2\beta\frac{d\theta}{dt} + \omega_0^2\theta = 0 \qquad \beta \equiv \frac{\gamma}{2m}.$$

$$\theta = e^{\lambda t}, \quad \lambda^2 + 2\beta\lambda + \omega_0^2 = 0, \quad \lambda = -\beta \pm \sqrt{\beta^2 - \omega_0^2}$$

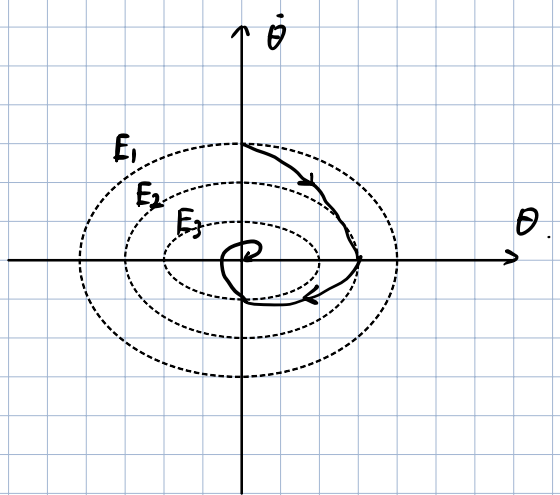
小阻尼, $\beta^2 - \omega_0^2 < 0$. $\lambda = -\beta \pm i\omega$, $\omega = \sqrt{\omega_0^2 - \beta^2}$

$$\theta = e^{-\beta t}(C_1 e^{i\omega t} + C_2 e^{-i\omega t})$$

阻尼

$$= pe^{-\beta t}\cos(\omega t + \varphi)$$

相空间? 吸引子.!



- 阻尼力使单摆闭合轨道破裂.
- 运动过程中依次穿过等能 E_1, E_2, E_3, …, 最终趋向原点.
- 这一类为最简单的吸引子, 零维.

- 鞍点仍位于 $\pm\Pi$ 处，将整个相平面分为 3 个区域. 以原点为吸引子的区域被称为该吸引子的**吸引域**.

杜芳方程

杜芳振子

$$\frac{d^2x}{dt^2} + \gamma\frac{dx}{dt} - k\cdot x + \zeta\cdot x^3 = 0$$

γ：阻尼系数　　R：振子的恢复力.

- $k<0,\ \zeta<0$ ：软弹簧振子.
- $k<0,\ \zeta>0$ ：硬弹簧振子.
- $k>0$ ：质点在双阱势中运动.

正弦驱动力下：

$$\frac{dx^2}{dt^2} + \gamma\frac{dx}{dt} - k\cdot x + \zeta x^3 = F\cos\nu t$$

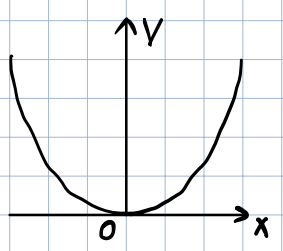
考虑无驱动无阻尼，$\zeta=1,\ \gamma=0$.

$$\frac{d^2x}{dt^2} - k\cdot x + x^3 = 0$$

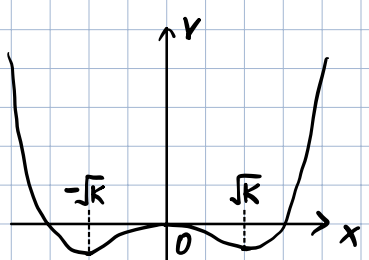
积分：

$$\underbrace{\frac{1}{2}\left(\frac{dx}{dt}\right)^2}_{\text{动能}K} - \underbrace{\frac{1}{2}\left(kx^2 - \frac{1}{2}x^4\right)}_{\text{势能}V} = \underbrace{E}_{\text{系统总能量}}.$$

$$V = \frac{1}{2}\left(\frac{1}{2}x^4 - k\cdot x^2\right).$$



$k<0$



$k>0$.

- $K<0$. 1个稳定平衡点，$x=0$.
- $K>0$. 3个平衡点：

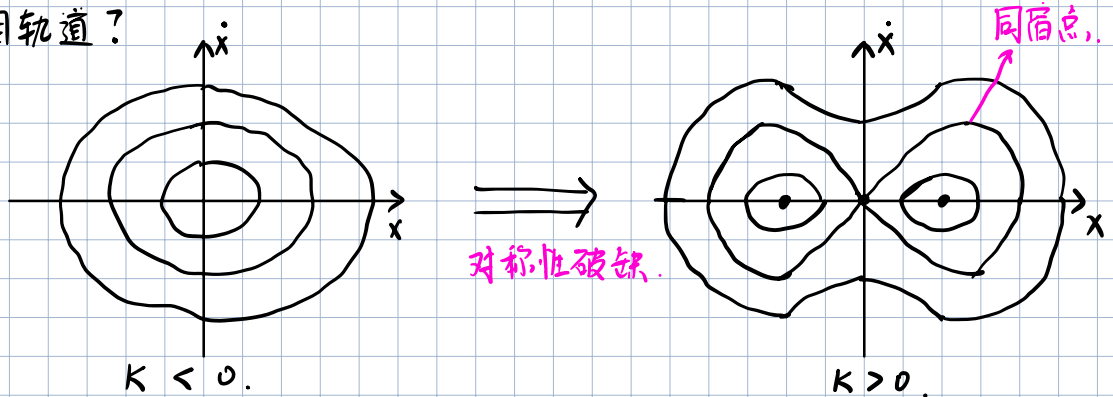
  $x=\pm\sqrt{K}$, 稳定平衡点.

  $x=0$ 鞍点.

杜芬方程的解在 $K=0$ 发生了分裂.

一个稳定解分裂为2个稳定解和1个不稳定解 —— 分岔.

相轨道？



非线性阻尼　范德波耳方程.

范德波尔方程.　可变非线性阻尼微分方程：

$$\frac{d^2x}{dt^2} + \varepsilon(x^2-1)\frac{dx}{dt} + \omega_0^2 x = 0.$$

可变非线性阻尼.

令其解为：

$$x = A\cos\omega t.$$

$$\frac{dx}{dt} = -A\omega\sin\omega t, \quad \frac{dx^2}{dt} = -A\omega^2\cos\omega t$$

代入：

$$(\omega_0^2-\omega^2)A\cos\omega t = -\varepsilon(A^2\cos^2\omega t - 1)(-A\omega\sin\omega t)$$

$$= \varepsilon A\omega\left(\frac{1}{4}A^2-1\right)\sin\omega t + \frac{1}{4}\varepsilon A^3\sin 3\omega t$$

两边同次谐波项系数应相等.　why ?

$$(\omega_0^2-\omega^2)A = 0,$$

$$\varepsilon A\omega\left(\frac{1}{4}A^2-1\right) = 0.$$

$\therefore$　$\omega = \omega_0$,　$A = A_c = 2$.　定常振幅.

右侧三次谐波项是一个小项可略去.

$$\varepsilon(x^2-1)\frac{dx}{dt} \approx \varepsilon\left(\frac{1}{4}A^2-1\right)\frac{dx}{dt}.$$

再代回原方程：

$$\frac{d^2x}{dt^2} + \varepsilon\left(\frac{1}{4}A^2-1\right)\frac{dx}{dt} + \omega_0^2 x = 0.$$

谐波线性化方程，形式上与阻尼单摆类似. 解：

$$x(t) = A_0 e^{\gamma t}\cos\omega t.$$

- γ：等价阻尼系数，与振幅相关：

$$\gamma = \gamma(A) = \frac{1}{2}\varepsilon\left(\frac{1}{4}A^2 - 1\right).$$

- ω：自振动频率：

$$\omega = \omega(A) = (\omega_0^2 - \gamma^2)^{1/2} = \left[\omega_0^2 - \frac{1}{4}\varepsilon^2\left(\frac{1}{4}A^2 - 1\right)^2\right]^{1/2}$$